Chapter 16

Timer Group (TIMG)

16.1 Overview

General-purpose timers can be used to precisely time an interval, trigger an interrupt after a particular interval (periodically and aperiodically), or act as a hardware clock. As shown in Figure 16.1-1, the ESP32-P4 chip contains two timer groups, namely timer group 0 and timer group 1 (hereinafter referred to as TIMG*n*, where *n* can be 0 or 1). Each timer group consists of two general-purpose timers (hereinafter referred to as T*x*, where *x* can be 0 or 1) and one Main System Watchdog Timer (MWDT). The general-purpose timer is based on a 16-bit prescaler and a 54-bit auto-reload-capable up-down counter.

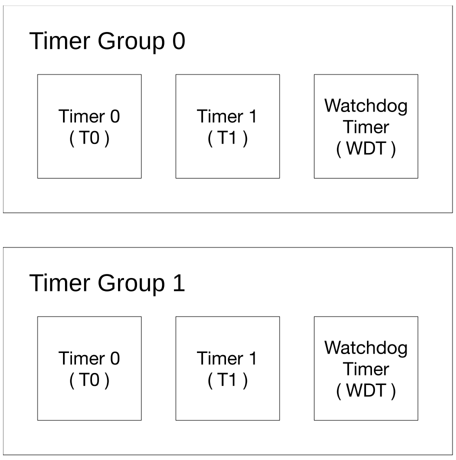

Figure 16.1-1. Timer Group Overview

Note that while the Main System Watchdog Timer registers are described in this chapter, their functional description is included in Chapter 17 *Watchdog Timers (WDT)*. Therefore, the term "timer" within this chapter refers to the general-purpose timer.

16.2 Features

The timer's features are summarized as follows:

- A 54-bit time-base counter programmable to incrementing or decrementing
- Three clock sources: PLL_F80M_CLK or XTAL_CLK or RC_FAST_CLK
- A 16-bit clock prescaler, from 2 to 65536
- Able to read real-time value of the time-base counter
- Able to halt and resume the time-base counter

- Programmable alarm generation
- Timer value reload — Auto-reload at alarm or software-controlled instant reload
- Calculate clock frequency — Calculate the measured frequency of the clock, which can be called TIMG0_CALI_CLK, based on the crystal clock
- Level interrupt generation
- Support several ETM tasks and events

16.3 Functional Description

Figure 16.3-1 shows Timer T*x* in timer group TIMG*n*. T*x* contains a 16-bit integer divider as a prescaler, a timer-based counter, and a comparator for alarm generation.

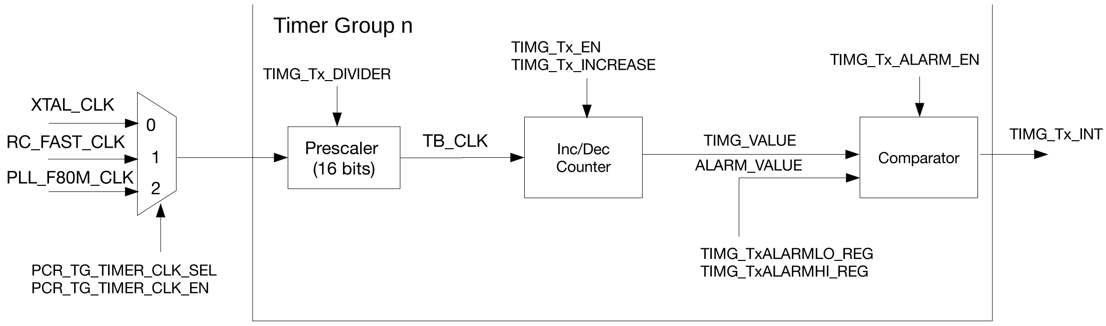

Figure 16.3-1. Timer Group Architecture

16.3.1 16-bit Prescaler and Clock Selection

Take the Timer T*x* in timer group TIMG*n* as an example:

- The timer can select its clock source by setting the HP_SYS_CLKRST_TIMERGRP*n*_T*x*_SRC_SEL field of the HP_SYS_CLKRST_PERI_CLK_CTRL20_REG register or the HP_SYS_CLKRST_PERI_CLK_CTRL21_REG register. When the field is 0, XTAL_CLK is selected; when the field is 1, RC_FAST_CLK is selected and when the field is 2, PLL_F80M_CLK is selected.
- The selected clock can be switched on by setting HP_SYS_CLKRST_TIMERGRP*n*_T*x*_CLK_EN field of the HP_SYS_CLKRST_PERI_CLK_CTRL20_REG register or the HP_SYS_CLKRST_PERI_CLK_CTRL21_REG register to 1 and switched off by setting it to 0. The clock is then divided by a 16-bit prescaler to generate the time-base counter clock (TB_CLK) used by the time-base counter. The divisor of the prescaler can be configured through the TIMG_T*x*_DIVIDER field.

TIMG_T*x*_DIVIDER field can be configured as 0 ~ 65535 for a divisor range of 2 ~ 65536. To be more specific, when TIMG_T*x*_DIVIDER is configured as:

- 0: the divisor is 65536
- 1: the divisor is 2
- 2: the divisor is also 2
- 3 ~ 65535: the divisor is 3 ~ 65535

To modify the 16-bit prescaler, please first configure the TIMG_Tx_DIVIDER field, and then set TIMG_Tx_DIVCNT_RST to 1. Meanwhile, the timer must be disabled (i.e., TIMG_Tx_EN should be cleared). Otherwise, the result can be unpredictable.

16.3.2 54-bit Time-base Counter

The 54-bit time-base counter is based on TB_CLK and can be configured to increment or decrement via the TIMG_Tx_INCREASE field. The time-base counter can be enabled or disabled by setting or clearing the TIMG_Tx_EN field, respectively. When enabled, the time-base counter increments or decrements on each cycle of TB_CLK. When disabled, the time-base counter is essentially frozen. Note that the TIMG_Tx_INCREASE field can be changed no matter whether TIMG_Tx_EN is set or not, and this will cause the time-base counter to change direction instantly.

To read the 54-bit value of the time-base counter, the timer value must be latched to two registers before being read by the CPU (due to the CPU being 32-bit). By writing any value to the TIMG_TxUPDATE_REG, the current value of the 54-bit timer starts to be latched into the TIMG_TxLO_REG and TIMG_TxHI_REG registers containing the lower 32-bits and higher 22-bits, respectively. When TIMG_TxUPDATE_REG is cleared by hardware, it indicates the latch operation has been completed and current timer value can be read from the TIMG_TxLO_REG and TIMG_TxHI_REG registers. TIMG_TxLO_REG and TIMG_TxHI_REG registers will remain unchanged for the CPU to read in its own time until TIMG_TxUPDATE_REG is written to again.

16.3.3 Alarm Generation

A timer can be configured to trigger an alarm when the timer's current value matches the alarm value. An alarm will cause an interrupt to occur and (optionally) an automatic reload of the timer's current value (see Section 16.3.4).

The 54-bit alarm value is configured using TIMG_TxALARMLO_REG and TIMG_TxALARMHI_REG, which represent the lower 32-bits and higher 22-bits of the alarm value, respectively. However, the configured alarm value is ineffective until the alarm is enabled by setting the TIMG_Tx_ALARM_EN field. To avoid alarm being enabled "too late" (i.e., the timer value has already passed the alarm value when the alarm is enabled), the hardware will trigger the alarm immediately if the current timer value is:

- higher than the alarm value (within a defined range) when the up-down counter increments
- lower than the alarm value (within a defined range) when the up-down counter decrements

Table 16.3-1 and Table 16.3-2 show the relationship between the current value of the timer, the alarm value, and when an alarm is triggered. The current time value and the alarm value are defined as follows:

- TIMG_VALUE = {TIMG_TxHI_REG, TIMG_TxLO_REG}
- ALARM_VALUE = {TIMG_TxALARMHI_REG, TIMG_TxALARMLO_REG}

Table 16.3-1. Alarm Generation When Up-Down Counter Increments

Scenario	Range	Alarm
1	ALARM_VALUE − TIMG_VALUE $> 2^{53}$	Triggered
2	0 $<$ ALARM_VALUE − TIMG_VALUE $\leq 2^{53}$	Triggered when the up-down counter counts TIMG_VALUE up to ALARM_VALUE
3	0 $\leq$ TIMG_VALUE − ALARM_VALUE $< 2^{53}$	Triggered
4	TIMG_VALUE − ALARM_VALUE $\geq 2^{53}$	Triggered when the up-down counter restarts counting up from 0 after reaching the timer's maximum value and counts TIMG_VALUE up to ALARM_VALUE

Table 16.3-2. Alarm Generation When Up-Down Counter Decrements

Scenario	Range	Alarm
5	TIMG_VALUE − ALARM_VALUE $> 2^{53}$	Triggered
6	0 $<$ TIMG_VALUE − ALARM_VALUE $\leq 2^{53}$	Triggered when the up-down counter counts TIMG_VALUE down to ALARM_VALUE
7	0 $\leq$ ALARM_VALUE − TIMG_VALUE $< 2^{53}$	Triggered
8	ALARM_VALUE − TIMG_VALUE $\geq 2^{53}$	Triggered when the up-down counter restarts counting down from the timer's maximum value after reaching the minimum value and counts TIMG_VALUE down to ALARM_VALUE

When an alarm occurs, the TIMG_Tx_ALARM_EN field is automatically cleared and no alarm will occur again until the TIMG_Tx_ALARM_EN is set next time.

16.3.4 Timer Reload

A timer is reloaded when a timer's current value is overwritten with a reload value stored in the TIMG_Tx_LOAD_LO and TIMG_Tx_LOAD_HI fields that correspond to the lower 32-bits and higher 22-bits of the timer's new value, respectively. However, writing a reload value to TIMG_Tx_LOAD_LO and TIMG_Tx_LOAD_HI will not cause the timer's current value to change. Instead, the reload value is ignored by the timer until a reload event occurs. A reload event can be triggered either by a software instant reload or an auto-reload at alarm.

A software instant reload is triggered by the CPU writing any value to TIMG_TxLOAD_REG, which causes the timer's current value to be instantly reloaded. If TIMG_Tx_EN is set, the timer will continue incrementing or decrementing from the new value. In this case if TIMG_Tx_ALARM_EN is set, the timer will still trigger alarms in scenarios listed in Table 16.3-1 and 16.3-2. If TIMG_Tx_EN is cleared, the timer will remain frozen at the new value until counting is re-enabled.

An auto-reload at alarm will cause a timer reload when an alarm occurs, thus allowing the timer to continue incrementing or decrementing from the reload value. This is generally useful for resetting the timer's value when using periodic alarms. To enable auto-reload at alarm, the TIMG_Tx_AUTORELOAD field should be set. If not enabled, the timer's value will continue to increment or decrement past the alarm value after an alarm.

16.3.5 Event Task Matrix Feature

The TIMG*n* on ESP32-P4 supports the Event Task Matrix (ETM) function, which allows TIMG*n*'s ETM tasks to be triggered by any peripherals' ETM events, or TIMG*n*'s ETM events to trigger any peripherals' ETM tasks. This section introduces the ETM tasks and events related to TIMG*n*. For more information, please refer to Chapter 13 Event Task Matrix (ETM).

TIMG*n* can receive the following ETM tasks:

- TG0_TASK_CNT_START_TIMER0: When triggered, it will enable the time-base counter.
- TG1_TASK_CNT_START_TIMER0: When triggered, it will enable the time-base counter.
- TG0_TASK_CNT_STOP_TIMER0: When triggered, it will disable the time-base counter.
- TG1_TASK_CNT_STOP_TIMER0: When triggered, it will disable the time-base counter.

> **Note:**
> The above two ETM tasks have the same function as the APB configuration TIMG_T0_EN. When these operations occur at the same time, the priority of each operation from high to low is as follows:
>
> 1. TG0_TASK_CNT_START_TIMER0 and TG1_TASK_CNT_START_TIMER0: When triggered, it will enable the time-base counter;
> 2. TG0_TASK_CNT_STOP_TIMER0 and TG1_TASK_CNT_STOP_TIMER0: When triggered, it will disable the time-base counter;
> 3. APB configuration TIMG_T0_EN: Enable or disable the time-base counter.

- TG0_TASK_ALARM_START_TIMER0: When triggered, it will enable the alarm generation.
- TG1_TASK_ALARM_START_TIMER0: When triggered, it will enable the alarm generation.

> **Note:**
> Alarm generation can also be enabled through APB method configuring TIMG_T0_ALARM_EN and hardware events. When these operations occur at the same time, the priority of each operation from high to low is as follows:
>
> 1. TG0_TASK_ALARM_START_TIMER0 and TG1_TASK_ALARM_START_TIMER0: When triggered, it will enable the alarm generation;
> 2. Alarm events: When triggered, it will disable the alarm generation;
> 3. APB configuration TIMG_T0_ALARM_EN: Enable or disable the alarm generation.

- TG0_TASK_CNT_CAP_TIMER0: When triggered, it will update the current counter value to the TIMG_T0LO_REG and TIMG_T0HI_REG registers.
- TG1_TASK_CNT_CAP_TIMER0: When triggered, it will update the current counter value to the TIMG_T0LO_REG and TIMG_T0HI_REG registers.
- TG0_TASK_CNT_RELOAD_TIMER0: When triggered, it will overwrite the current counter value with the reload value stored in TIMG_T0_LOAD_LO and TIMG_T0_LOAD_HI.
- TG1_TASK_CNT_RELOAD_TIMER0: When triggered, it will overwrite the current counter value with the

reload value stored in TIMG_T0_LOAD_LO and TIMG_T0_LOAD_HI.

TIMG*n* can generate the following ETM events:

- TG0_EVT_CNT_CMP_TIMER0: Indicates the interrupt event of T0 in TIMG0.
- TG1_EVT_CNT_CMP_TIMER0: Indicates the interrupt event of T0 in TIMG1.

> **Note:**
> All the ETM tasks and events of TIMG*n* will not take effect until the TIMG*n*_ETM_EN is set to 1.

In practical applications, timer groups' ETM events can trigger their own ETM tasks.

For example, TG0_TASK_ALARM_START_TIMER0 and TG1_TASK_ALARM_START_TIMER0 can be triggered respectively by TG0_EVT_CNT_CMP_TIMER0 and TG1_EVT_CNT_CMP_TIMER0 to realize periodic alarm. For configuration steps, please refer to 16.4.4 Timer as Periodic Alarm by ETM.

16.3.6 RTC_SLOW_CLK Frequency Calculation

Using XTAL_CLK as a reference, it is possible to calculate the frequency of clock sources for RTC_SLOW_CLK (i.e., RC_SLOW_CLK, RC_FAST_DIV_CLK, and XTAL32K_CLK) as follows:

1. Start periodic or one-shot frequency calculation (see Section 16.4.5 for details);
2. Once receiving the signal to start calculation, the counter of XTAL_CLK and the counter of RTC_SLOW_CLK begin to work at the same time. When the counter of RTC_SLOW_CLK counts to C0, the two counters stop counting simultaneously;
3. Assume the value of XTAL_CLK's counter is C1, and the frequency of RTC_SLOW_CLK would be calculated as: $f\_rtc = \frac{C0 \times f\_XTAL\_CLK}{C1}$

16.3.7 Interrupts

ESP32-P4's TIMG*n* can generate the following interrupt signal(s) that will be sent to the Interrupt Matrix.

- TG*n*_T*x*_INT
- TG*n*_WDT_INT

There are several internal interrupt sources from TIMG*n* that can generate the above interrupt signal(s). The interrupt sources from TIMG*n* are listed with their trigger conditions and the resulted interrupt signal(s) in Table 16.3-3.

Table 16.3-3. TIMG*n*'s Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
TIMG_T*x*_INT	Alarm generated by T*x*	TG*n*_T*x*_INT
TIMG_WDT_INT	Alarm generated by WDT	TG*n*_WDT_INT

> **Note:**
> For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter 12 *Interrupt Matrix* > Section 12.2 *Interrupt Terminology in ESP32-P4*.

Each interrupt source can be configured by a common set of registers that are described in Section *Interrupt Configuration Registers*. The specific registers can be found in Section 16.5 *Register Summary*.

16.4 Configuration and Usage

16.4.1 Timer as a Simple Clock

1. Configure the time-base counter
   - Select clock source by configuring the HP_SYS_CLKRST_TIMERGRP0_TGRT_CLK_SRC_SEL field of the HP_SYS_CLKRST_PERI_CLK_CTRL21_REG register.
   - Configure the 16-bit prescaler by setting TIMG_Tx_DIVIDER.
   - Configure the timer direction by setting or clearing TIMG_Tx_INCREASE.
   - Set the timer's starting value by writing the starting value to TIMG_Tx_LOAD_LO and TIMG_Tx_LOAD_HI, then reloading it into the timer by writing any value to TIMG_TxLOAD_REG.
2. Start the timer by setting TIMG_Tx_EN.
3. Get the timer's current value.
   - Write any value to TIMG_TxUPDATE_REG to latch the timer's current value.
   - Wait until TIMG_TxUPDATE_REG is cleared by hardware.
   - Read the latched timer value from TIMG_TxLO_REG and TIMG_TxHI_REG.

16.4.2 Timer as One-shot Alarm

1. Configure the time-base counter following step 1 of Section 16.4.1.
2. Configure the alarm.
   - Configure the alarm value by setting TIMG_TxALARMLO_REG and TIMG_TxALARMHI_REG.
   - Enable interrupt by setting TIMG_Tx_INT_ENA.
3. Disable auto reload by clearing TIMG_Tx_AUTORELOAD.
4. Start the alarm by setting TIMG_Tx_ALARM_EN.
5. Handle the alarm interrupt.
   - Clear the interrupt by setting the timer's corresponding bit in TIMG_Tx_INT_CLR.
   - Disable the timer by clearing TIMG_Tx_EN.

16.4.3 Timer as Periodic Alarm by APB

1. Configure the time-base counter following step 1 in Section 16.4.1.
2. Configure the alarm following step 2 in Section 16.4.2.
3. Enable auto reload by setting TIMG_Tx_AUTORELOAD and configure the reload value via TIMG_Tx_LOAD_LO and TIMG_Tx_LOAD_HI.

4. Start the alarm by setting TIMG_Tx_ALARM_EN.
5. Handle the alarm interrupt (repeat on each alarm iteration).
   - Clear the interrupt by setting the timer's corresponding bit in TIMG_Tx_INT_CLR.
   - If the next alarm requires a new alarm value and reload value (i.e., different alarm interval per iteration), then TIMG_TxALARMLO_REG, TIMG_TxALARMHI_REG, TIMG_Tx_LOAD_LO, and TIMG_Tx_LOAD_HI should be reconfigured as needed. Otherwise, the aforementioned registers should remain unchanged.
   - Re-enable the alarm by setting TIMG_Tx_ALARM_EN.
6. Stop the timer (on final alarm iteration).
   - Clear the interrupt by setting the timer's corresponding bit in TIMG_Tx_INT_CLR.
   - Disable the timer by clearing TIMG_Tx_EN.

16.4.4 Timer as Periodic Alarm by ETM

1. Enable the ETM module's clock
2. Map ETM event to ETM task (which means using the event to trigger the task)
   - If TIMG_T0_AUTORELOAD is set to 1, map TG0_EVT_CNT_CMP_TIMER0 and TG1_EVT_CNT_CMP_TIMER0 to the TG0_TASK_ALARM_START_TIMER0 and TG1_TASK_ALARM_START_TIMER0 respectively by one ETM channel.
   - If TIMG_T0_AUTORELOAD is set to 0, in addition to mapping TG0_EVT_CNT_CMP_TIMER0 and TG1_EVT_CNT_CMP_TIMER0 to the TG0_TASK_ALARM_START_TIMER0 and TG1_TASK_ALARM_START_TIMER0, the TG0_EVT_CNT_CMP_TIMER0 and TG1_EVT_CNT_CMP_TIMER0 should also be mapped to TG0_TASK_CNT_RELOAD_TIMER0 and TG1_TASK_CNT_RELOAD_TIMER0 by another ETM channel.
3. Choose to enable the one or two ETM channels.
4. Set TIMER_ETM_EN to 1 to enable timer group's ETM events and tasks.
5. Configure the time-base counter following step 1 in Section 16.4.1.
6. Configure the alarm following step 2 in Section 16.4.2.
7. Configure the reload value via TIMG_T0_LOAD_LO and TIMG_T0_LOAD_HI.
8. Handle the TG0_EVT_CNT_CMP_TIMER0 and TG1_EVT_CNT_CMP_TIMER0.
   - When alarm generates, the TG0_EVT_CNT_CMP_TIMER0 and TG1_EVT_CNT_CMP_TIMER0 also generate, and the alarm generation will be disabled by the alarm.
   - If TIMG_T0_AUTORELOAD is 1, the current counter value is overwritten by the reloaded value. The alarm generation will be reopened by TG0_TASK_ALARM_START_TIMER0 and TG1_TASK_ALARM_START_TIMER0.
   - If TIMG_T0_AUTORELOAD is 0, the current counter value is overwritten by the reloaded value because of the TG0_TASK_CNT_RELOAD_TIMER0 and TG1_TASK_CNT_RELOAD_TIMER0. The alarm generation will be reopened by TG0_TASK_ALARM_START_TIMER0 and TG1_TASK_ALARM_START_TIMER0.

9. Stop the timer (on final alarm iteration).
   - Disable the ETM channels used to map timer group's event and task
   - Set TIMER_ETM_EN to 0.
   - Clear the interrupt by setting the timer's corresponding bit in TIMG_T0_INT_CLR.
   - Disable the timer by clearing TIMG_T0_EN.

16.4.5 RTC_SLOW_CLK Frequency Calculation

1. One-shot frequency calculation
   - Configure HP_SYS_CLKRST_TIMERGRP0_TGRT_CLK_EN to enable clock gate of TIMG0_CALI_CLK.
   - Configure HP_SYS_CLKRST_TIMERGRP0_TGRT_CLK_SRC_SEL to select one clock as the clock source of TIMG0_CALI_CLK.
   - Configure HP_SYS_CLKRST_TIMERGRP0_TGRT_CLK_DIV_NUM to output TIMG0_CALI_CLK after integer prescaler.
   - Configure the time of calculation via TIMG_RTC_CALI_MAX.
   - Select one-shot frequency calculation by clearing TIMG_RTC_CALI_START_CYCLING, and enable the two counters via TIMG_RTC_CALI_START.
   - Once TIMG_RTC_CALI_RDY becomes 1, read TIMG_RTC_CALI_VALUE to get the value of XTAL_CLK's counter, and calculate the frequency of RTC_SLOW_CLK according to the formula in Section 16.3.6.
2. Periodic frequency calculation
   - Select the clock whose frequency is to be calculated (clock source of RTC_SLOW_CLK) via HP_SYS_CLKRST_TIMERGRP0_TGRT_CLK_SRC_SEL, and configure the time of calculation via TIMG_RTC_CALI_MAX.
   - Select periodic frequency calculation by enabling TIMG_RTC_CALI_START_CYCLING.
   - When TIMG_RTC_CALI_CYCLING_DATA_VLD is 1, TIMG_RTC_CALI_VALUE is valid.
3. Timeout

   If the counter of RTC_SLOW_CLK cannot finish counting in TIMG_RTC_CALI_TIMEOUT_RST_CNT cycles, TIMG_RTC_CALI_TIMEOUT will be set to indicate a timeout.

16.5 Register Summary

The addresses in this section are relative to Timer Group base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
T0 Control and configuration registers			
TIMG_T0CONFIG_REG	Timer 0 configuration register	0x0000	varies
TIMG_T0LO_REG	Timer 0 current value, low 32 bits	0x0004	RO
TIMG_T0HI_REG	Timer 0 current value, high 22 bits	0x0008	RO
TIMG_T0UPDATE_REG	Write to copy current timer value to TIMG_T0LO_REG or TIMG_T0HI_REG	0x000C	R/W/SC
TIMG_T0ALARMLO_REG	Timer 0 alarm value, low 32 bits	0x0010	R/W
TIMG_T0ALARMHI_REG	Timer 0 alarm value, high 22 bits	0x0014	R/W
TIMG_T0LOADLO_REG	Timer 0 reload value, low 32 bits	0x0018	R/W
TIMG_T0LOADHI_REG	Timer 0 reload value, high 22 bits	0x001C	R/W
TIMG_T0LOAD_REG	Write to reload timer from TIMG_T0LOADLO_REG or TIMG_T0LOADHI_REG	0x0020	WT
T1 Control and configuration registers			
TIMG_T1CONFIG_REG	Timer 1 configuration register	0x0024	varies
TIMG_T1LO_REG	Timer 1 current value, low 32 bits	0x0028	RO
TIMG_T1HI_REG	Timer 1 current value, high 22 bits	0x002C	RO
TIMG_T1UPDATE_REG	Write to copy current timer value to TIMG_T1LO_REG or TIMG_T1HI_REG	0x0030	R/W/SC
TIMG_T1ALARMLO_REG	Timer 1 alarm value, low 32 bits	0x0034	R/W
TIMG_T1ALARMHI_REG	Timer 1 alarm value, high 22 bits	0x0038	R/W
TIMG_T1LOADLO_REG	Timer 1 reload value, low 32 bits	0x003C	R/W
TIMG_T1LOADHI_REG	Timer 1 reload value, high 22 bits	0x0040	R/W
TIMG_T1LOAD_REG	Write to reload timer from TIMG_T1LOADLO_REG or TIMG_T1LOADHI_REG	0x0044	WT
WDT Control and configuration registers			
TIMG_WDTCONFIG0_REG	Watchdog timer configuration register	0x0048	varies
TIMG_WDTCONFIG1_REG	Watchdog timer prescaler register	0x004C	varies
TIMG_WDTCONFIG2_REG	Watchdog timer stage 0 timeout value	0x0050	R/W
TIMG_WDTCONFIG3_REG	Watchdog timer stage 1 timeout value	0x0054	R/W
TIMG_WDTCONFIG4_REG	Watchdog timer stage 2 timeout value	0x0058	R/W
TIMG_WDTCONFIG5_REG	Watchdog timer stage 3 timeout value	0x005C	R/W
TIMG_WDTFEED_REG	Write to feed the watchdog timer	0x0060	WT
TIMG_WDTWPROTECT_REG	Watchdog write protect register	0x0064	R/W
RTC CALI Control and configuration registers			
TIMG_RTCCALICFG_REG	RTC calibration configure register	0x0068	varies
TIMG_RTCCALICFG1_REG	RTC calibration configure register 1	0x006C	RO
TIMG_RTCCALICFG2_REG	RTC calibration configure register 2	0x0080	varies

Name	Description	Address	Access
Interrupt registers			
TIMG_INT_ENA_TIMERS_REG	Interrupt enable bits	0x0070	R/W
TIMG_INT_RAW_TIMERS_REG	Raw interrupt status	0x0074	R/SS/WTC
TIMG_INT_ST_TIMERS_REG	Masked interrupt status	0x0078	RO
TIMG_INT_CLR_TIMERS_REG	Interrupt clear bits	0x007C	WT
Version register			
TIMG_NTIMERS_DATE_REG	Timer version control register	0x00F8	R/W
Clock configuration registers			
TIMG_REGCLK_REG	Timer group clock gate register	0x00FC	R/W

16.6 Registers

The addresses in this section are relative to Timer Group base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section Programming Reserved Register Field.

Register 16.1. TIMG_TxCONFIG_REG (x: 0-1) (0x0000+0x24*x)

TIMG_Tx_EN	TIMG_Tx_INCREASE	TIMG_Tx_AUTORELOAD	TIMG_Tx_DIVIDER	TIMG_Tx_DIVCNT_RST	(reserved)	TIMG_Tx_ALARM_EN	(reserved)	
31	30	29	28 … 13	12	11	10	9 … 0	
0	1	1	0x01	0	0	0	0 0 0 0 0 0 0 0 0 0	Reset

TIMG_Tx_ALARM_EN Configures whether to enable Timer Tx alarm function. This bit will be automatically cleared once an alarm occurs.
0: Disable
1: Enable
(R/W/SC)

TIMG_Tx_DIVCNT_RST Configures to reset Timer Tx's clock divider counter.
0: No effect
1: Reset
(WT)

TIMG_Tx_DIVIDER Represents Timer x clock (Tx_clk) prescaler value. (R/W)

TIMG_Tx_AUTORELOAD Configures to enable Timer Tx auto-reload function at the time of alarm.
0: No effect
1: Enable
(R/W)

TIMG_Tx_INCREASE Configures the counting direction of Timer Tx time-base counter.
0: Decrement
1: Increment
(R/W)

TIMG_Tx_EN Configures whether to enable Timer Tx time-base counter.
0: Disable
1: Enable
(R/W/SS/SC)

Register 16.2. TIMG_TxLO_REG (x: 0-1) (0x0004+0x24*x)

TIMG_Tx_LO	
31 ... 0	
0x000000	Reset

TIMG_Tx_LO Represents the low 32 bits of the time-base counter of Timer Tx. Valid only after writing to TIMG_TxUPDATE_REG.
Measurement unit: Tx_clk.
(RO)

Register 16.3. TIMG_TxHI_REG (x: 0-1) (0x0008+0x24*x)

(reserved)	TIMG_TO_HI	
31 ... 22	21 ... 0	
0 0 0 0 0 0 0 0 0 0	0x0000	Reset

TIMG_Tx_HI Represents the high 22 bits of the time-base counter of Timer Tx. Valid only after writing to TIMG_TxUPDATE_REG.
Measurement unit: Tx_clk.
(RO)

Register 16.4. TIMG_TxUPDATE_REG (x: 0-1) (0x000C+0x24*x)

TIMG_Tx_UPDATE	(reserved)	
31	30 ... 0	
0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	Reset

TIMG_Tx_UPDATE Configures to latch the counter value.
0: Latch
1: Latch
(R/W/SC)

Register 16.5. TIMG_TxALARMLO_REG (x: 0-1) (0x0010+0x24*x)

TIMG_Tx_ALARM_LO	
31 ... 0	
0x000000	Reset

TIMG_Tx_ALARM_LO Configures the low 32 bits of Timer Tx alarm trigger time-base counter value. Valid only when TIMG_Tx_ALARM_EN is 1.
Measurement unit: Tx_clk.
(R/W)

Register 16.6. TIMG_TxALARMHI_REG (x: 0-1) (0x0014+0x24*x)

(reserved)	TIMG_Tx_ALARM_HI	
31 ... 22	21 ... 0	
0 0 0 0 0 0 0 0 0 0	0x0000	Reset

TIMG_Tx_ALARM_HI Configures the high 22 bits of Timer Tx alarm trigger time-base counter value. Valid only when TIMG_Tx_ALARM_EN is 1.
Measurement unit: Tx_clk.
(R/W)

Register 16.7. TIMG_TxLOADLO_REG (x: 0-1) (0x0018+0x24*x)

TIMG_Tx_LOAD_LO	
31 ... 0	
0x000000	Reset

TIMG_Tx_LOAD_LO Configures low 32 bits of the value that a reload will load onto Timer Tx time-base counter.
Measurement unit: Tx_clk.
(R/W)

Register 16.8. TIMG_TxLOADHI_REG (x: 0-1) (0x001C+0x24*x)

(reserved)	TIMG_Tx_LOAD_HI	
31 ... 22	21 ... 0	
0 0 0 0 0 0 0 0 0 0	0x0000	Reset

TIMG_Tx_LOAD_HI Configures high 22 bits of the value that a reload will load onto Timer Tx time-base counter.
Measurement unit: Tx_clk.
(R/W)

Register 16.9. TIMG_TxLOAD_REG (x: 0-1) (0x0020+0x24*x)

TIMG_Tx_LOAD	
31 ... 0	
0x000000	Reset

TIMG_Tx_LOAD Write any value to trigger Timer Tx time-base counter reload. (WT)

Register 16.10. TIMG_WDTCONFIG0_REG (0x0048)

TIMG_WDT_EN	TIMG_WDT_STG0	TIMG_WDT_STG1	TIMG_WDT_STG2	TIMG_WDT_STG3	TIMG_WDT_CONF_UPDATE_EN	(reserved)	TIMG_WDT_CPU_RESET_LENGTH	TIMG_WDT_SYS_RESET_LENGTH	TIMG_WDT_FLASHBOOT_MOD_EN	TIMG_WDT_PROCPU_RESET_EN	TIMG_WDT_APPCPU_RESET_EN	(reserved)	
31	30 29	28 27	26 25	24 23	22	21	20 18	17 15	14	13	12	11 0	
0	0	0	0	0	0	0	0x1	0x1	1	0	0	0 0 0 0 0 0 0 0 0 0 0 0	Reset

TIMG_WDT_APPCPU_RESET_EN Configures whether to mask the APP CPU reset generated by MWDT. Valid only when write protection is disabled.

0: Mask

1: Unmask

(R/W)

TIMG_WDT_PROCPU_RESET_EN Configures whether to mask the PRO CPU reset generated by MWDT. Valid only when write protection is disabled.

0: Mask

1: Unmask

(R/W)

TIMG_WDT_FLASHBOOT_MOD_EN Configures whether to enable flash boot protection.

0: Disable

1: Enable

(R/W)

TIMG_WDT_SYS_RESET_LENGTH Configures the system reset signal length. Valid only when write protection is disabled.

Measurement unit: mwdt_clk.

0: 100 ns	4: 500 ns
1: 200 ns	5: 800 ns
2: 300 ns	6: 1.6 μs
3: 400 ns	7: 3.2 μs

(R/W)

Continued on the next page...

Register 16.10. TIMG_WDTCONFIG0_REG (0x0048)

Continued from the previous page...

TIMG_WDT_CPU_RESET_LENGTH Configures the CPU reset signal length. Valid only when write protection is disabled.

Measurement unit: mwdt_clk.

0: 100 ns	4: 500 ns
1: 200 ns	5: 800 ns
2: 300 ns	6: 1.6 μs
3: 400 ns	7: 3.2 μs

(R/W)

TIMG_WDT_CONF_UPDATE_EN Configures to update the WDT configuration registers.

0: No effect
1: Update
(WT)

TIMG_WDT_STG3 Configures the timeout action of stage 3. Valid only when write protection is disabled.

0: No effect
1: Interrupt
2: Reset CPU
3: Reset system
(R/W)

TIMG_WDT_STG2 Configures the timeout action of stage 2. Valid only when write protection is disabled.

0: No effect
1: Interrupt
2: Reset CPU
3: Reset system
(R/W)

TIMG_WDT_STG1 Configures the timeout action of stage 1. Valid only when write protection is disabled.

0: No effect
1: Interrupt
2: Reset CPU
3: Reset system
(R/W)

Continued on the next page...

Register 16.10. TIMG_WDTCONFIG0_REG (0x0048)

Continued from the previous page...

TIMG_WDT_STG0 Configures the timeout action of stage 0. Valid only when write protection is disabled.
0: No effect
1: Interrupt
2: Reset CPU
3: Reset system
(R/W)

TIMG_WDT_EN Configures whether or not to enable the MWDT. Valid only when write protection is disabled.
0: Disable
1: Enable
(R/W)

Register 16.11. TIMG_WDTCONFIG1_REG (0x004C)

TIMG_WDT_CLK_PRESCALE	(reserved)	TIMG_WDT_DIVCNT_RST	
31 16	15 1	0	
0x01	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

TIMG_WDT_DIVCNT_RST Configures whether to reset WDT's clock divider counter.
 0: No effect
 1: Reset
 (WT)

TIMG_WDT_CLK_PRESCALE Configures MWDT clock prescaler value. Valid only when write protection is disabled.
 MWDT clock period = 12.5 ns *TIMG_WDT_CLK_PRESCALE. (R/W)

Register 16.12. TIMG_WDTCONFIG2_REG (0x0050)

TIMG_WDT_STG0_HOLD	
31 0	
26000000	Reset

TIMG_WDT_STG0_HOLD Configures the stage 0 timeout value. Valid only when write protection is disabled.
 Measurement unit: mwdt_clk.
 (R/W)

Register 16.13. TIMG_WDTCONFIG3_REG (0x0054)

TIMG_WDT_STG1_HOLD	
31 … 0	
0x7ffffff	Reset

TIMG_WDT_STG1_HOLD Configures the stage 1 timeout value. Valid only when write protection is disabled.
Measurement unit: mwdt_clk.
(R/W)

Register 16.14. TIMG_WDTCONFIG4_REG (0x0058)

TIMG_WDT_STG2_HOLD	
31 … 0	
0x0fffff	Reset

TIMG_WDT_STG2_HOLD Configures the stage 2 timeout value. Valid only when write protection is disabled.
Measurement unit: mwdt_clk.
(R/W)

Register 16.15. TIMG_WDTCONFIG5_REG (0x005C)

TIMG_WDT_STG3_HOLD	
31 … 0	
0x0fffff	Reset

TIMG_WDT_STG3_HOLD Configures the stage 3 timeout value. Valid only when write protection is disabled.
Measurement unit: mwdt_clk.
(R/W)

Register 16.16. TIMG_WDTFEED_REG (0x0060)

TIMG_WDT_FEED	
31 0	
0x000000	Reset

TIMG_WDT_FEED Write any value to feed the MWDT. Valid only when write protection is disabled. (WT)

Register 16.17. TIMG_WDTWPROTECT_REG (0x0064)

TIMG_WDT_WKEY	
31 0	
0x50d83aa1	Reset

TIMG_WDT_WKEY Configures a different value than its reset value to enable write protection. (R/W)

Register 16.18. TIMG_RTCCALICFG_REG (0x0068)

TIMG_RTC_CALI_START	TIMG_RTC_CALI_MAX	TIMG_RTC_CALI_RDY	(reserved)	TIMG_RTC_CALI_START_CYCLING	(reserved)	
31	30 … 16	15	14 … 13	12	11 … 0	
0	0x01	0	0 0	1	0 0 0 0 0 0 0 0 0 0 0 0	Reset

TIMG_RTC_CALI_START_CYCLING Configures the frequency calculation mode.
0: One-shot frequency calculation
1: Periodic frequency calculation.
(R/W)

TIMG_RTC_CALI_RDY Represents whether one-shot frequency calculation is done.
0: Not done
1: Done
(RO)

TIMG_RTC_CALI_MAX Configures the time to calculate RTC slow clock's frequency.
Measurement unit: XTAL_CLK.
(R/W)

TIMG_RTC_CALI_START Configures whether to enable one-shot frequency calculation.
0: Disable
1: Enable
(R/W)

Register 16.19. TIMG_RTCCALICFG1_REG (0x006C)

TIMG_RTC_CALI_VALUE	(reserved)	TIMG_RTC_CALI_CYCLING_DATA_VLD	
31 … 7	6 … 1	0	
0x00000	0 0 0 0 0 0	0	Reset

TIMG_RTC_CALI_CYCLING_DATA_VLD Represents whether periodic frequency calculation is done.
0: Not done
1: Done
(RO)

TIMG_RTC_CALI_VALUE Represents the value countered by XTAL_CLK when one-shot or periodic frequency calculation is done. It is used to calculate RTC slow clock's frequency. (RO)

Register 16.20. TIMG_RTCCALICFG2_REG (0x0080)

TIMG_RTC_CALI_TIMEOUT_THRES	TIMG_RTC_CALI_TIMEOUT_RST_CNT	(reserved)	TIMG_RTC_CALI_TIMEOUT	
31 … 7	6 … 3	2 … 1	0	
0x1ffffff	3	0 0	0	Reset

TIMG_RTC_CALI_TIMEOUT Represents whether RTC frequency calculation is timeout.
0: No timeout
1: Timeout
(RO)

TIMG_RTC_CALI_TIMEOUT_RST_CNT Configures the cycles that reset frequency calculation timeout.
Measurement unit: XTAL_CLK.
(R/W)

TIMG_RTC_CALI_TIMEOUT_THRES Configures the threshold value for the RTC frequency calculation timer. If the timer's value exceeds this threshold, a timeout is triggered.
Measurement unit: XTAL_CLK.
(R/W)

Register 16.21. TIMG_INT_ENA_TIMERS_REG (0x0070)

Bits	31–3	2	1	0	
Field	(reserved)	TIMG_WDT_INT_ENA	TIMG_T1_INT_ENA	TIMG_T0_INT_ENA	
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	Reset

TIMG_Tx_INT_ENA (x: 0-1) Write 1 to enable the TIMG_Tx_INT interrupt. (R/W)

TIMG_WDT_INT_ENA Write 1 to enable the TIMG_WDT_INT interrupt. (R/W)

Register 16.22. TIMG_INT_RAW_TIMERS_REG (0x0074)

Bits	31–3	2	1	0	
Field	(reserved)	TIMG_WDT_INT_RAW	TIMG_T1_INT_RAW	TIMG_T0_INT_RAW	
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	Reset

TIMG_Tx_INT_RAW (x: 0-1) The raw interrupt status bit of the TIMG_Tx_INT interrupt. (R/SS/WTC)

TIMG_WDT_INT_RAW The raw interrupt status bit of the TIMG_WDT_INT interrupt. (R/SS/WTC)

Register 16.23. TIMG_INT_ST_TIMERS_REG (0x0078)

Bits	31–3	2	1	0	
Field	(reserved)	TIMG_WDT_INT_ST	TIMG_T1_INT_ST	TIMG_T0_INT_ST	
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	Reset

TIMG_Tx_INT_ST (x: 0-1) The masked interrupt status bit of the TIMG_Tx_INT interrupt. (RO)

TIMG_WDT_INT_ST The masked interrupt status bit of the TIMG_WDT_INT interrupt. (RO)

Register 16.24. TIMG_INT_CLR_TIMERS_REG (0x007C)

Bits	31–3	2	1	0	
Field	(reserved)	TIMG_WDT_INT_CLR	TIMG_T1_INT_CLR	TIMG_T0_INT_CLR	
Reset value	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	Reset

TIMG_Tx_INT_CLR (*x*: 0-1) Write 1 to clear the TIMG_Tx_INT interrupt. (WT)

TIMG_WDT_INT_CLR Write 1 to clear the TIMG_WDT_INT interrupt. (WT)

Register 16.25. TIMG_NTIMERS_DATE_REG (0x00F8)

Bits	31–28	27–0	
Field	(reserved)	TIMG_NTIMGS_DATE	
Reset value	0 0 0 0	0x2209142	Reset

TIMG_NTIMGS_DATE Version control register (R/W)

Register 16.26. TIMG_REGCLK_REG (0x00FC)

Bits	31	30–29	28	27–0	
Field	TIMG_CLK_EN	(reserved)	TIMG_ETM_EN	(reserved)	
Reset value	0	0 0	1	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	Reset

TIMG_ETM_EN Configures whether to enable timer's ETM task and event.
- 0: Disable
- 1: Enable
- (R/W)

TIMG_CLK_EN Configures whether to enable gate clock signal for registers.
- 0: Force clock on for registers
- 1: Support clock only when registers are read or written to by software.
- (R/W)